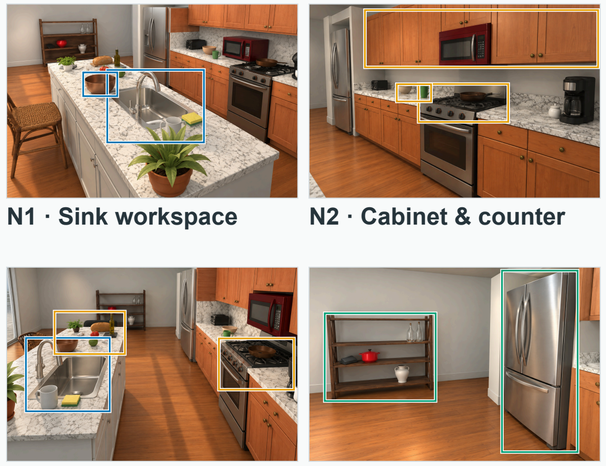
CAFE

Scene-grounded reasoning with three functional zones

Spatial evidence links task opportunities to grounded actions through Trigger-LLM and TRI-LLM.

Observed scene evidence

NODE OBSERVATIONS



N1 · Sink workspace

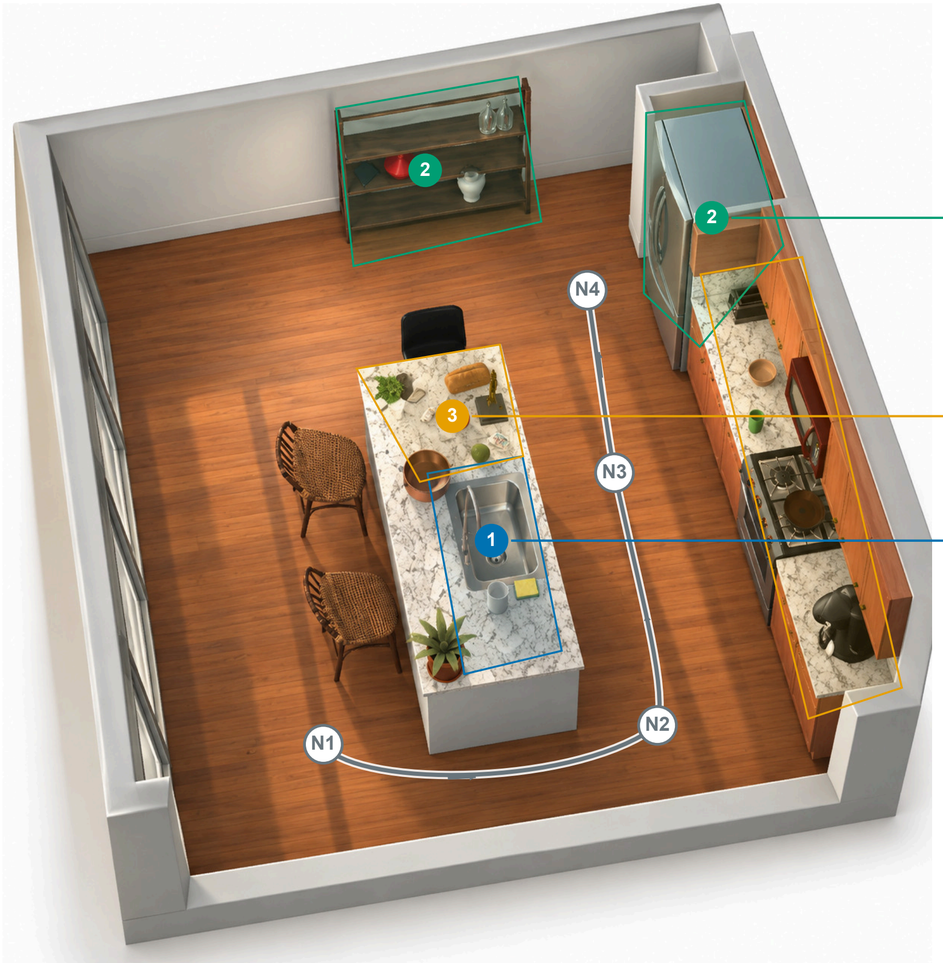
N2 · Cabinet & counter

N3 · Island & stove

N4 · Fridge & shelf

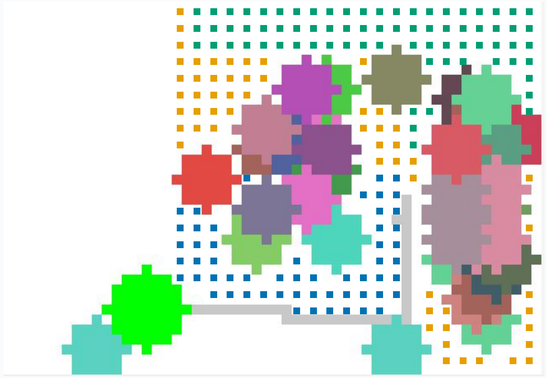
Illustrative views from each observation node.

AI2-THOR scene reconstruction



Three-zone semantic map

ORIGINAL SEMANTIC MAP



2 **Storage**
Fridge + wooden shelf

3 **Mix**
Prep / worktops

1 **Washing**
Sink / tools

OBJECT COLOR KEY

- Dish sponge
- Drawer
- Faucet
- Floor
- Fork
- Houseplant
- Microwave
- Mug
- Pan
- Paper towel roll
- Pepper shaker
- Plate
- Pot
- Potato
- Salt shaker
- Sink
- Sink basin
- Soap bottle
- Spatula
- Spoon
- Stove burner

Trigger-LLM

Compare → de-duplicate → count

≥ 4 distinct task candidates
OR ≥ 20 new objects (fallback)

NO TRIGGER — Explore; retain reference

TRIGGER — Admit TRI-LLM reasoning

Trigger

TRI-LLM

01 Draft
Propose scene tasks
Wash bowl
Store potato
Grounded task proposals

02 Complete
Bind objects / states
Sink · cloth · fridge
Targets + geometry
Completed parameters

03 Audit
Verify dependencies
Fridge: open → put
On → clean → off
Executable sequence

SCENE-LEVEL EXECUTION

01 Locate bowl → 02 Pick up bowl → 03 Locate cloth → 04 Turn on faucet → 05 Wash bowl → 06 Place bowl → 07 Pick potato → 08 Go to fridge → 09 Put potato